

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 06:21 UTC

SEC P vs NP — notebook summary

Generated 2026-05-18 06:21 UTC

- Cycles recorded: **769**
- Time span: 584.7h (~1.32 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	729
BARRIER_HIT	20
FALSIFIED	16
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	40
Schur-Weyl Duality	28
Free Probability Theory	19
Representation Theory of Symmetric Groups	18
Matroid Theory	16
Free Probability	14
Additive Combinatorics	13
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Persistent Homology	5
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4

Field	Cycles
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-17 20:47 UTC	INCONCLUSIVE	Fekete Capacity of Laplacian Spectrum Upper-Bounds
2026-05-17 21:20 UTC	INCONCLUSIVE	Max-Cut SoS-2
2026-05-17 21:53 UTC	INCONCLUSIVE	Polymer Cluster-Expansion
2026-05-17 22:25 UTC	FALSIFIED	Convergence Radius Bounds
2026-05-17 22:43 UTC	INCONCLUSIVE	Tree-Res Size
2026-05-17 23:19 UTC	INCONCLUSIVE	Fatgraph Genus of Canonical
2026-05-17 23:46 UTC	INCONCLUSIVE	Clause Ribbon Lower-Bounds
2026-05-18 00:33 UTC	INCONCLUSIVE	Resolution
2026-05-18 03:29 UTC	INCONCLUSIVE	Forman-Ricci Min-Curvature of
2026-05-18 03:58 UTC	INCONCLUSIVE	Term-Overlap Graph
2026-05-18 04:25 UTC	INCONCLUSIVE	Lower-Bounds Mon
		Bourgain-Tzafriri Sub-Column
		Spectral Excess Bounds DISJ
		CC_R
		Trivial-Isotype Mass Gap
		Separates Perm from Padded
		Det
		Eulerian Descent Entropy
		Separates Perm from Padded
		Det_m
		Signed-Laplacian Holonomy
		Spectrum Bounds Tseitin
		Tree-Resolution
		Bonami-Beckner 4-2
		Saturation of Clause-Sum
		Bounds DPLL Depth
		Cycle-Type Mixing Defect of
		Barrington BP Bounds NC^1
		Length
		NW-Design Restriction
		Dimension Separates Perm
		from Padded Det

Time	Verdict	Title
2026-05-18 04:47 UTC	INCONCLUSIVE	Coxeter Inversion Spread of Barrington BPs Bounds NC^1 Depth
2026-05-18 05:16 UTC	INCONCLUSIVE	Birkhoff Spectral Gap of Gate Co-occurrence Bounds AC^0 PARITY Co
2026-05-18 05:56 UTC	INCONCLUSIVE	Mansour L1 Fourier Mass of Clause-Falsification Polynomial Bounds
2026-05-18 06:21 UTC	INCONCLUSIVE	Chang Spectrum Dimension of Wire-Source DFS Labels Bounds $ACC^0[2]$

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```